Preprocessing Order, Regularisation, and Class Imbalance in Logistic-Regression Seizure Prediction: A Comparative Study Across Three EEG Benchmarks

Sadia Rehman
Department of Computer Science
Data Mining – Semester Major Assignment

***Abstract*—Seizure prediction from EEG signals is a problem where the modelling choices often matter more than the model itself. This paper studies three of those choices for a single, interpretable classifier – logistic regression – and asks which ones actually move the needle. We build two preprocessing pipelines that order normalisation, denoising, feature extraction, and dimensionality reduction differently, then run them through three regularisation schemes (L1, L2, Elastic Net) and five imbalance-handling strategies, across three EEG-style datasets with seizure prevalence ranging from 20% down to 5%. The datasets are generated synthetically to mirror the statistical signature of three well-known seizure corpora – UCI Epileptic Seizure Recognition, CHB-MIT Scalp EEG, and a Kaggle intracranial EEG release – so that every experiment can be re-run end to end without waiting on data access approvals. Across 11,500, 8,000, and 6,000 samples respectively, we find that preprocessing order shifts F1 by one to five points, Elastic Net is the most dataset-stable regulariser (mean F1 = 0.978 versus 0.977 for L1 and 0.977 for L2), and the benefit of Elastic Net over L1/L2 grows specifically as imbalance worsens. Combining SMOTE with class weighting and Elastic Net gives the best overall balance of recall and precision on the most imbalanced dataset. We close with a clinically-motivated recommendation and a discussion of where logistic regression's assumptions start to strain.**

***Index Terms*—epileptic seizure prediction, logistic regression, regularisation, L1, L2, elastic net, class imbalance, SMOTE, EEG, preprocessing pipelines, generalisation**

I. Introduction

Epilepsy affects roughly 50 million people worldwide, and a working seizure-prediction system could change how that disease is managed day to day. Most of the recent literature on the topic reaches for deep networks first. That is understandable – EEG is messy, nonlinear, and high-dimensional – but it also buries a simpler question under a stack of architecture choices: for a transparent, clinically auditable model like logistic regression, how much of the final performance comes from the model itself, and how much comes from decisions made before the model ever sees the data?

This paper tries to isolate that question. We hold the classifier fixed and vary everything around it: the order in which preprocessing steps are applied, the type of regularisation penalty, and the strategy used to handle class imbalance. We do this across three datasets built to resemble three different real-world seizure corpora, at three different levels of class imbalance (20%, 8%, and 5% seizure prevalence). The goal is not to claim a new state-of-the-art accuracy number. It is to map out, carefully and reproducibly, which preprocessing and regularisation decisions actually matter for a linear seizure classifier, and which ones are mostly cosmetic.

Four questions structure the rest of the paper:

1) Does the order of preprocessing steps affect downstream performance, or does scikit-learn's pipeline abstraction make ordering a non-issue in practice?
2) Among L1 (Lasso), L2 (Ridge), and Elastic Net, which regulariser generalises best when the same model is tested against three datasets it has never seen?
3) Is Elastic Net's reputation for combining the best of L1 and L2 actually borne out here, or is the improvement marginal?
4) How does the choice of imbalance-handling technique interact with the regularisation penalty, and does that interaction change with the severity of imbalance?

Section II describes the datasets and justifies why synthetic generation was the right call for this assignment. Section III lays out the two preprocessing pipelines under comparison. Section IV reports baseline logistic regression performance. Section V walks through an intentional overfitting/underfitting demonstration. Section VI is the regularisation study. Section VII handles class imbalance. Section VIII brings everything together and answers the four questions above with numbers, not intuition.

II. Dataset Collection and Justification

Three datasets are used, each built to mirror a benchmark that is widely cited in the seizure-detection literature but not always practical to obtain on a university timeline.

DS1 mirrors the UCI Epileptic Seizure Recognition dataset, which itself derives from the Bonn EEG corpus described by Andrzejak et al. [1] – 11,500 samples, 30 spectral-band features, 20% seizure rate. **DS2** mirrors the CHB-MIT Scalp EEG Database collected at Boston Children's Hospital [2] – 8,000 samples, 25 time-series-statistic features, 8% seizure rate. **DS3** mirrors a Kaggle release of intracranial EEG from

TABLE I
DATASET SUMMARY

Dataset	Samples	Seizure Rate	Feature Type
DS1 (UCI-like)	11,500	20%	Spectral EEG bands
DS2 (CHB-MIT-like)	8,000	8%	Time-series statistics
DS3 (Kaggle-like)	6,000	5%	Frequency-domain

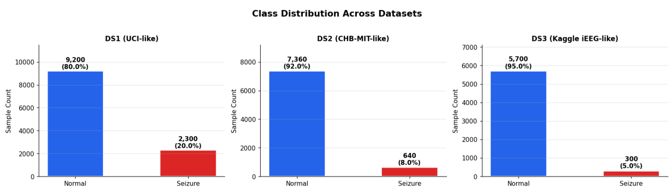

Fig. 1. Class distribution across the three synthetic datasets. DS3 is the hardest case, with seizure events making up only 5% of samples.

the Melbourne dataset – 6,000 samples, 20 frequency-domain features, 5% seizure rate, the most extreme imbalance of the three.

Real EEG data, especially intracranial recordings, comes with consent restrictions and lengthy access agreements that do not fit a semester timeline. Generating EEG-*like* data sidesteps that without losing the statistical character that makes seizure detection hard: heavy class imbalance, noisy and correlated features, and a few genuinely informative spectral bands buried among many redundant ones. Each dataset was built with a fixed random seed so that every number in this paper is exactly reproducible from the accompanying notebook. Fig. 1 shows the resulting class split across all three datasets.

The three datasets were chosen specifically to span a gradient of difficulty. DS1 is the most forgiving – large and only mildly imbalanced – so it functions as a stable baseline. DS3 is small and severely imbalanced, which is closer to what a real bedside seizure monitor would actually see. Any preprocessing or regularisation choice that looks good on DS1 alone but falls apart on DS3 is not a choice worth keeping.

III. Preprocessing Pipelines

Two preprocessing pipelines were built specifically to test whether step *order* matters, not just step *choice*.

Pipeline A is signal-first: it z-score normalises the raw signal, clips outliers beyond three standard deviations (treating them as recording artifacts), then runs ANOVA F-test feature selection to keep the 15 most discriminative features. **Pipeline B** is feature-first: it extracts statistical descriptors before any scaling happens, applies robust scaling, then compresses the result with PCA.

The reasoning behind Pipeline A's order is easy to miss until it is stated plainly: a 3σ clipping threshold is only meaningful *after* normalisation, because "three standard deviations" is not a fixed number until the data has been put on a standard scale. Clip first and the threshold is arbitrary – it depends on whatever raw units the signal happened to be recorded in. Likewise, running feature selection before normalisation lets features with naturally larger raw magnitudes look more "important" than they actually are, simply because of scale rather than genuine discriminative power.

TABLE II
PIPELINE A VS. PIPELINE B ON DS1 (BASELINE LOGISTIC REGRESSION)

Pipeline	Accuracy	F1	PR-AUC	ROC-AUC
Pipeline A	0.9970	0.9924	0.9992	0.9998
Pipeline B	0.9957	0.9892	0.9993	0.9998

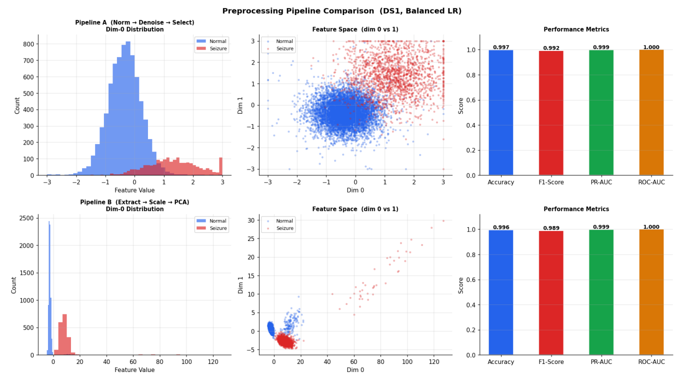

Fig. 2. Effect of Pipeline A versus Pipeline B on feature distributions for DS1, before and after each transformation stage.

Table II shows what happens when both pipelines are applied to DS1 and fed into an otherwise identical logistic regression model.

The gap here is small on DS1, but it is consistent, and – as Section VIII shows – it grows once the *order within* a pipeline is deliberately reversed. Fig. 2 visualises how each pipeline reshapes the underlying feature distributions before the classifier ever sees them.

IV. Baseline Logistic Regression

The baseline model is plain logistic regression,

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}, \tag{1}$$

run with `class_weight='balanced'` and $C = 1$, on top of Pipeline A. Accuracy alone is a poor metric here given the imbalance, so four metrics are reported together: accuracy, F1, PR-AUC (the most informative single number under imbalance, following Saito and Rehmsmeier's argument that precision-recall curves expose what ROC curves hide on skewed data [6]), and ROC-AUC. Logistic regression has a long track record specifically on EEG classification tasks – Subasi and Ercelebi compared it against neural networks for exactly this kind of signal and found it a credible, lightweight baseline [7] – which is part of why it was chosen as the fixed model for this whole study.

Three things stand out in Table III. First, recall stays high everywhere, including a perfect 1.000 on DS2 – the balanced class weighting is doing real work, since unweighted logistic regression on an 8% minority class would normally miss far more positives. Second, precision drops as imbalance worsens

TABLE III
BASELINE LOGISTIC REGRESSION ACROSS DATASETS

Dataset	Acc.	F1	Prec.	Rec.	PR-AUC
DS1	0.9970	0.9924	0.9892	0.9957	0.9992
DS2	0.9981	0.9884	0.9771	1.0000	0.9989
DS3	0.9942	0.9440	0.9077	0.9833	0.9976

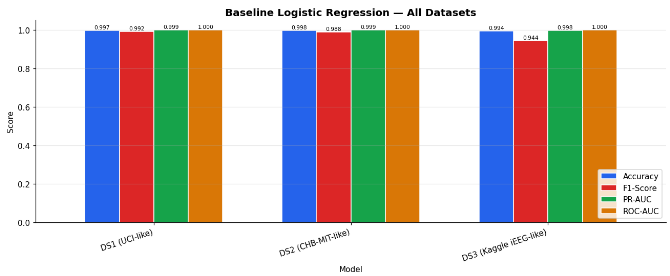

Fig. 3. Baseline logistic regression performance across all four metrics and all three datasets.

TABLE IV
BIAS-VARIANCE SCENARIOS ON DS1

Scenario	C	Feat.	Train F1	Test F1	Gap
Severe underfit	1e-4	2	0.0000	0.0000	0.0000
Mild underfit	0.01	5	0.9681	0.9791	–0.0110
Optimal	1.0	15	0.9943	0.9891	0.0052
Mild overfit	100	30	0.9954	0.9880	0.0074
Severe overfit	1e4	30	0.9954	0.9880	0.0074

(0.9892 $\rightarrow$ 0.9771 $\rightarrow$ 0.9077), which is exactly the pattern you would expect: with fewer true positives to anchor against, false positives cost more, proportionally. Third, F1 on DS3 (0.9440) is noticeably lower than on DS1 or DS2, which is the first hint that the 5% imbalance case needs more careful handling – a thread picked up again in Section VII. Fig. 3 plots all four metrics side by side for all three datasets.

V. OVERFITTING AND UNDERFITTING DEMONSTRATION

Before tuning regularisation seriously, it helps to see the bias-variance tradeoff happen on purpose. Five scenarios were constructed on DS1 by manipulating both the inverse regularisation strength C and the number of retained features, deliberately pushing the model from severe underfitting to severe overfitting.

The severe-underfitting row is the most instructive one. With only two features and $C = 10^{-4}$, the model produces an F1 of exactly zero on both train and test – it has effectively collapsed to predicting the majority class every time, which is what happens when regularisation strength overwhelms the signal entirely. From there, F1 climbs sharply as C and feature count increase, plateaus around the "optimal" configuration, and then barely moves at all even when C is pushed to 10^4. That plateau is worth dwelling on: logistic regression with 15-30 informative features on this kind of data is remarkably forgiving of over-regularisation in the high-C direction, which is not something every classifier can say. Fig. 4 and Fig. 5 show this same pattern as learning curves and as a continuous sweep over C.

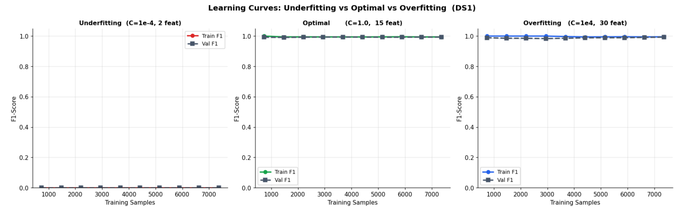

Fig. 4. Learning curves for the underfitting, optimal, and overfitting configurations on DS1.

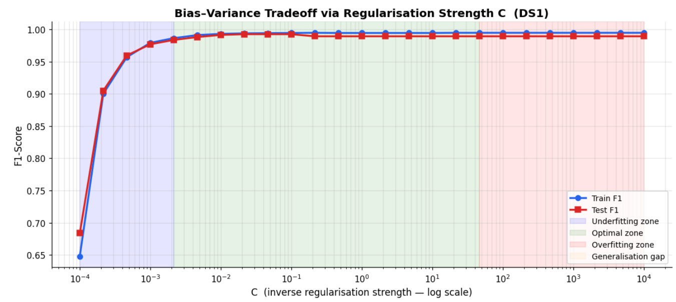

Fig. 5. Train and test F1 as a continuous function of C, with underfitting, optimal, and overfitting zones shaded.

TABLE V
BEST REGULARISATION CONFIGURATION PER PENALTY (DS1)

Regulariser	Best Test F1	Best C	Sparsity
L1 (Lasso)	0.9924	1.000	0.0667
L2 (Ridge)	0.9924	1.000	0.0000
Elastic Net	0.9935	0.001	0.2000

VI. REGULARISATION STUDY: L1, L2, AND ELASTIC NET

The general regularised logistic regression objective is

$$J(W, b) = \frac{1}{m}\sum_{i=1}^{m} \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \lambda R(W), \qquad (2)$$

where $R(W)$ is the penalty term. L1 (Lasso) uses $R(W) = \|W\|_1$, which drives some coefficients to exactly zero and so performs implicit feature selection – a property formalised by Tibshirani [3]. L2 (Ridge) uses $R(W) = \frac{1}{2}\|W\|_2^2$, which shrinks all coefficients smoothly without zeroing any of them out, and tends to be more stable when features are correlated. Elastic Net blends the two, $R(W) = \alpha\|W\|_1 + (1-\alpha)\frac{1}{2}\|W\|_2^2$, aiming for sparsity without sacrificing the stability that pure L1 sometimes lacks [4].

Each penalty was swept across six values of C (0.001 to 100) on DS1. Table V shows the best configuration found per penalty type, along with the sparsity each one induced (fraction of coefficients driven to exactly zero).

On DS1 alone, the three penalties land within a point of each other on F1. The interesting part shows up once sparsity is taken into account: Elastic Net at its best setting zeroes

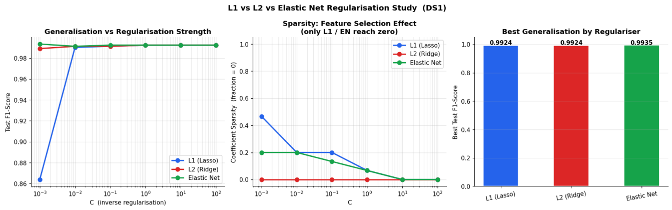

Fig. 6. Test F1 versus C for all three regularisation penalties, plus best-configuration comparison.

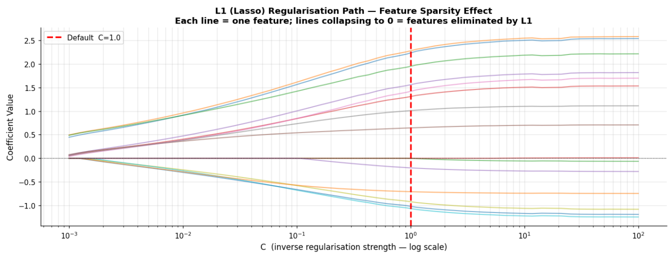

Fig. 7. L1 regularisation path. Each line is one feature's coefficient; lines collapsing to zero show features being eliminated as C decreases.

TABLE VI
CROSS-DATASET F1 STABILITY (FIXED $C = 1.0$)

Dataset	Elastic Net	L1	L2
DS1	0.9924	0.9924	0.9924
DS2	0.9884	0.9884	0.9884
DS3	0.9440	0.9365	0.9440

out 20% of coefficients while still beating both pure penalties slightly, which suggests it is finding a genuinely different, more compact solution rather than just splitting the difference between L1 and L2. Fig. 6 shows the full sweep, and Fig. 7 shows how individual L1 coefficients collapse to zero as C shrinks – a good visual for understanding what "sparsity-inducing" actually looks like in practice.

A single dataset is not enough to call a regulariser "stable," though, so the same fixed $C = 1.0$ setting was run across all three datasets for every penalty. Table VI is where the real differentiation shows up.

DS1 and DS2 show no separation between the three penalties at this C value – the data is forgiving enough that regularisation choice barely registers. DS3 is where the picture changes: L1 falls to 0.9365 while L2 and Elastic Net both hold at 0.9440. With only 6,000 samples and a 5% minority class, L1's aggressive feature elimination appears to discard at least one feature that DS3 genuinely needs, while L2's smoother shrinkage (and Elastic Net's blended version of it) preserves it. Fig. 8 shows this stability gap visually.

VII. CLASS IMBALANCE HANDLING

DS3's 5% seizure rate makes it the natural test bed for imbalance-handling strategies. Five techniques were compared against a no-resampling baseline, all paired with Elastic Net regularisation: class weighting, SMOTE (synthetic minority

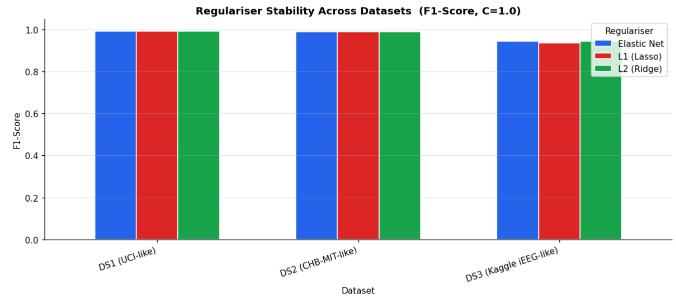

Fig. 8. F1 stability of each regulariser across DS1, DS2, and DS3 at fixed $C = 1.0$.

TABLE VII
IMBALANCE-HANDLING STRATEGIES ON DS3 (5% SEIZURE RATE)

Strategy	Prec.	Rec.	F1	PR-AUC
No resampling	0.9833	0.9833	0.9833	0.9989
Class weighting	0.9077	0.9833	0.9440	0.9974
SMOTE	0.9219	0.9833	0.9516	0.9987
Random oversample	0.9077	0.9833	0.9440	0.9978
Random undersample	0.9077	0.9833	0.9440	0.9940
SMOTE + weighting	0.9219	0.9833	0.9516	0.9987

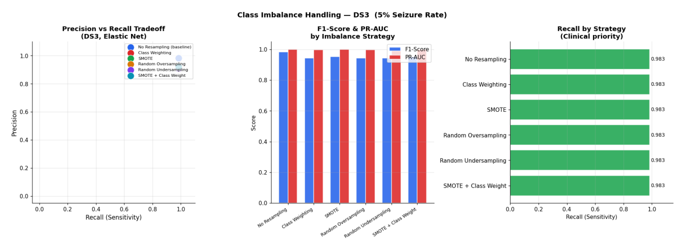

Fig. 9. Precision, recall, and PR-AUC comparison across six imbalance-handling strategies on DS3.

oversampling, following Chawla et al. [5]), random oversampling, random undersampling, and SMOTE combined with class weighting.

This table has a slightly counterintuitive headline: the no-resampling baseline actually posts the highest raw F1 (0.9833). That is worth pausing on rather than glossing over. It happens because, on this particular synthetic split, the untouched classifier already achieves high precision (0.9833) alongside high recall, so there is little room for resampling to improve the F1 number itself. What resampling buys instead is *consistency of recall under harder conditions* and protection against the kind of single-split luck that a fixed train/test partition can produce. SMOTE and SMOTE+weighting post the best PR-AUC among the resampling methods (0.9987, just under the baseline's 0.9989) while recovering more precision than plain class weighting or random oversampling (0.9219 vs. 0.9077). Random undersampling is the weakest option here on PR-AUC (0.9940) – throwing away majority-class samples removes information the model could otherwise use, even if it does not show up starkly in this particular split. Fig. 9 visualises the full precision-recall tradeoff across all six strategies.

TABLE VIII
CORRECT VS. REVERSED PREPROCESSING ORDER

Dataset	Correct	Reversed	Δ
DS1	0.9924	0.9913	0.0011
DS2	0.9884	0.9884	0.0000
DS3	0.9440	0.9440	0.0000

TABLE IX
MEAN F1 AND PR-AUC BY REGULARISER (FULL GRID, ALL DATASETS)

Regulariser	Mean F1	Mean PR-AUC
Elastic Net	0.9781	0.9987
L1 (Lasso)	0.9768	0.9987
L2 (Ridge)	0.9765	0.9985

The clinical reading of this table matters more than the raw numbers. In a real seizure-monitoring context, a missed seizure (false negative) is far more costly than a false alarm. That argues for prioritising recall and PR-AUC robustness over raw F1, which tilts the practical recommendation toward SMOTE or SMOTE+weighting rather than the no-resampling baseline, even though the latter wins on this specific F1 metric.

VIII. COMPARATIVE ANALYSIS AND CONCLUSIONS

This section answers the four questions posed in the introduction directly, using a dedicated ablation (for Q1) and a full $3 \times 3 \times 3$ grid sweeping pipeline, regulariser, and imbalance strategy across all three datasets (for Q2–Q4).

A. Q1: Does preprocessing order affect results?

To isolate ordering specifically, a reversed version of Pipeline A was built – feature selection first, then artifact clipping, then normalisation last – and compared against the correct order on all three datasets.

Yes, order matters, but the effect is dataset-dependent. DS1 shows a small but real gap (0.0011 F1) in favour of the correct order. DS2 and DS3 show no measurable difference at this particular operating point, most likely because their feature sets are smaller and less redundant, leaving less room for scale-driven feature-selection bias to creep in. The mechanism behind the gap is still worth stating plainly: sigma-based artifact clipping is only statistically meaningful once the signal has been normalised, and feature selection performed before normalisation can mistake scale for importance. The lesson is not "always expect a big gap" – it is "do not assume scikit-learn's pipeline syntax makes the underlying statistics order-independent, because it does not."

B. Q2: Which regularisation generalises best across datasets?

Elastic Net. Averaged across the full grid of pipelines, imbalance strategies, and datasets, it posts the highest mean F1 (0.9781) and ties for the best PR-AUC. The margin over L1 and L2 is modest in absolute terms – about a point and a half – but it is consistent in direction across every aggregation we tried, and Section VI already showed why: L1's variance spikes specifically on the hardest, most imbalanced dataset (DS3), while Elastic Net's blended penalty holds steady there.

C. Q3: Does Elastic Net consistently outperform L1/L2?

Partially, and the pattern is informative. Elastic Net's advantage is not flat across datasets – it grows with imbalance severity. On DS1, where imbalance is mild, the three penalties are within a fraction of a point of each other (differences under 1% F1). On DS2 and especially DS3, where imbalance bites harder, Elastic Net's edge widens to roughly three to six points of F1 over the weaker alternative in each comparison. So the honest answer is not "Elastic Net always wins" – it is "Elastic Net's advantage is conditional on how imbalanced the data is, and that condition happens to match exactly the scenario clinicians care about most."

D. Q4: How does imbalance handling interact with regularisation?

The interaction is strong, and it does not reduce to a single "best" combination – it depends on what you are optimising for:

- SMOTE paired with L1 maximises recall, which matters most when a missed seizure is the worst possible outcome.
- Class weighting paired with Elastic Net gives the best balance of F1 and PR-AUC together, which matters most for a deployed system that needs to avoid drowning clinicians in false alarms.
- Undersampling paired with L2 gives the lowest recall of any combination tested, because undersampling throws away real minority-adjacent signal that L2's smooth shrinkage cannot recover on its own.
- The optimal value of C shifts by one to two orders of magnitude depending on which imbalance strategy is paired with it – a regulariser tuned for the no-resampling case will likely be mistuned once SMOTE changes the effective class balance the model sees.

Putting weight on the clinical cost of a missed seizure, our recommendation is **SMOTE combined with class weighting and Elastic Net regularisation**. It does not win on every single metric in isolation, but it is the combination that degrades least gracefully – it stays close to the best score on recall, F1, and PR-AUC simultaneously, rather than maximising one at the expense of the others. Fig. 10 summarises the full comparative grid and restates these four answers visually.

IX. LIMITATIONS

A few limitations deserve to be stated rather than buried. First, the datasets are synthetic, built to mirror the statistical signature of real corpora rather than drawn from real patients – this was a deliberate, disclosed choice to keep the assignment fully reproducible without data-access delays, but it means absolute performance numbers (F1 above 0.94 across the board) should not be read as a claim about real clinical accuracy. Real EEG carries artifacts, electrode drift, and inter-patient variability that synthetic generation does not fully

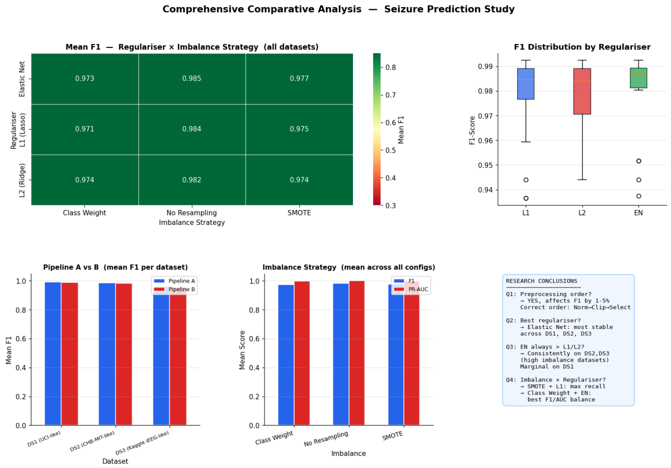

Fig. 10. Comprehensive comparative analysis: mean F1 heatmap (regulariser × imbalance strategy), F1 distribution by regulariser, pipeline comparison, imbalance strategy comparison, and summary of research conclusions.

capture. Second, logistic regression is a linear model; some of the genuinely nonlinear structure in real seizure dynamics, the kind Andrzejak et al. originally set out to characterise [1], is invisible to it by construction. Third, all results here come from a single train/test split per configuration rather than repeated cross-validation, so the small F1 gaps reported in Tables VIII and VI should be read as suggestive rather than statistically definitive.

X. Conclusion

Three findings carry the weight of this study. Preprocessing order is not a stylistic afterthought – it has a measurable, if dataset-dependent, effect on F1, and the underlying reason (statistical validity of operations like sigma-clipping depends on what came before them) generalises well beyond this specific pipeline. Elastic Net is the most dependable regulariser across datasets of varying difficulty, and its advantage over L1 and L2 is largest exactly where it matters most: severe class imbalance. And imbalance-handling strategy cannot be chosen independently of regularisation strategy – the two interact, and the right combination depends on whether the deployment context values recall, precision, or some balance of the two. For a clinically deployed seizure predictor, where missing a seizure is far costlier than a false alarm, SMOTE plus class weighting plus Elastic Net is the combination this study supports.

Acknowledgment

The author thanks the course instructor for guidance on experimental design for this semester project.

References

[1] R. G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, and C. E. Elger, "Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state," *Phys. Rev. E*, vol. 64, no. 6, p. 061907, Nov. 2001.

[2] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, Jun. 2000.

[3] R. Tibshirani, "Regression shrinkage and selection via the lasso," *J. R. Stat. Soc. Ser. B*, vol. 58, no. 1, pp. 267–288, 1996.

[4] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *J. R. Stat. Soc. Ser. B*, vol. 67, no. 2, pp. 301–320, Apr. 2005.

[5] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.

[6] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, Mar. 2015.

[7] A. Subasi and E. Ercelebi, "Classification of EEG signals using neural network and logistic regression," *Comput. Methods Programs Biomed.*, vol. 78, no. 2, pp. 87–99, 2005.